

Business Plan for Decision Tree Assignment

SmartLoan Partners: Small Business Loan Pre-Approval

Teaching case for Decision Tree classification, model evaluation, feature importance, and business interpretation

Business Case Element	Description
Company name	SmartLoan Partners
Business area	Financial services / small business lending
Main service	Screening small business loan applications and supporting pre-approval decisions.
Machine learning task	Binary classification using a Decision Tree model.
Target variable	Loan Approved
Prediction goal	Predict whether a small business loan application should be approved or not approved.

1. Business Overview

SmartLoan Partners is a small financial technology company that helps small and medium-sized businesses apply for business loans. The company collects information about each applicant, including business age, annual revenue, profit, credit score, existing debt, collateral, cash-flow stability, and previous default history.

The goal of the business is to make the loan screening process faster, more consistent, and more data-driven while still keeping human review in important financial decisions.

2. Business Problem

Loan officers receive many applications every month. Reviewing each application manually takes time, and different officers may make decisions using slightly different judgment. The company wants to use historical application data to identify which applicants are likely to be approved.

Machine learning can help the business create an initial decision-support score. This score does not replace loan officers, but it can help prioritize applications, identify risky cases, and support more consistent screening.

3. Decision That Machine Learning Can Support

The Decision Tree model can help SmartLoan Partners decide whether a loan application should be marked as likely approved or likely not approved. This can support business decisions such as:

- Which applications should be reviewed first by loan officers.
- Which applications may need additional documentation or collateral.
- Which applicants may be too risky for automatic pre-approval.
- Which factors are most important when evaluating small business loan applications.

4. Dataset Description

The related Excel dataset contains synthetic small business loan application records. Each row represents one loan application. The dataset includes both numerical and categorical features, and it intentionally includes a small number of missing values and duplicate rows so that students can practice data cleaning.

Variable Group	Examples	Business Meaning
Financial features	Annual_Revenue, Monthly_Profit, Existing_Debt, Requested_Loan_Amount	Show the financial strength and repayment capacity of the applicant.
Risk features	Credit_Score, Previous_Default, Cash_Flow_Stability	Help estimate whether the applicant may become a risky borrower.
Relationship features	Relationship_Length_Months, Online_Application_Complete	Show whether the applicant has an existing relationship and completed the process properly.
Business profile features	Industry, Region, Business_Age_Years	Describe the applicant's business context.
Target variable	Loan_Approved	The classification label that the Decision Tree model should predict.

5. Target Variable and Input Features

Target variable: Loan_Approved

Loan_Approved is the variable students should predict. It has two possible values: Yes and No. Yes means the application is approved. No means the application is not approved.

Input features:

- Industry and Region
- Business_Age_Years
- Annual_Revenue and Monthly_Profit
- Existing_Debt and Requested_Loan_Amount
- Loan_Term_Months and Collateral_Value
- Credit_Score and Previous_Default
- Online_Application_Complete
- Relationship_Length_Months
- Cash_Flow_Stability

6. Why This Prediction Is Useful for the Business

This prediction is useful because it can help SmartLoan Partners reduce manual screening time, identify high-risk applications earlier, and focus human attention on cases that require deeper review. It can also help the business understand which application features are most strongly related to approval decisions.

7. Business Interpretation Guide for Students

After training the Decision Tree model, students should interpret the model results using business language, not only technical language. They should connect accuracy, precision, recall, F1-score, false positives, false negatives, and feature importance to the lending business problem.

Model Result	Business Interpretation
High accuracy	The model correctly classifies many applications, but

	students should still check precision and recall.
False positive	The model predicts approval, but the application should not be approved. This may create financial loss if a risky borrower receives a loan.
False negative	The model predicts rejection, but the application could have been approved. This may cause lost revenue and unfair rejection of a good applicant.
Precision	Important because the business wants to avoid approving risky applications.
Recall	Important if the business wants to avoid missing good applicants who deserve approval.
Feature importance	Shows which variables the Decision Tree used most when making predictions, such as credit score, debt, profit, or collateral.

8. Possible Limitation or Bias

A possible limitation is that the dataset is synthetic and simplified. Real loan decisions may involve more complex financial records, legal requirements, market conditions, and ethical considerations. The model may also learn bias if historical approval decisions were unfair or if some industries or regions were overrepresented or underrepresented in the data.

9. Why Human Judgment Is Still Needed

Human judgment is still needed because loan approval affects real businesses and people. A Decision Tree can support the decision, but it cannot fully understand every context, exception, or ethical issue. Loan officers should review important cases, check documents, consider fairness, and make the final decision.

10. Suggested Student Submission Focus

Students should submit a notebook and GitHub repository that include data loading, data inspection, data cleaning, feature/target separation, preprocessing, train-test split, Decision Tree training, evaluation metrics, training/testing accuracy comparison, feature importance, and a short business interpretation.